Description

Illustration





Figure caption

1. Close-up of the material. © John Fernandez 2. Georgia State Capitol. © John Fernandez

The material

Limestone is formed of petrified sea-creatures. In its pure form it is chalk (when porous) or marble (when fully dense). Most limestone is not pure but contains sand (silicate) and other minerals. The most interesting limestones contain visible shells and other fossils of the creatures from which they derived. It is easy to cut; where it is available it is widely used as a building stone.

Compositional summary

Calcium carbonate (CaCO3) plus silicate impurities.

General properties

Density	2.55e3	-	2.6e3	kg/m^3
Price	* 0.31	-	0.785	GBP/kg
Date first used	-10000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✓

Mechanical properties

Young's modulus	35	-	55	GPa

Shear modulus		14	-	22	GPa
Bulk modulus		23	-	35	GPa
Bending modulus	*	20	-	50	GPa
Poisson's ratio		0.2	-	0.26	
Yield strength (elastic limit)		8	-	22	MPa
Tensile strength		8	-	22	MPa
Compressive strength	*	30	-	200	MPa
Bending strength	*	3	-	11	MPa
Elongation	*	0.01	-	0.06	% strain
Hardness - Vickers		3	-	18	HV
Fatigue strength at 10^7 cycles	*	4	-	16	MPa
Fracture toughness		0.6	-	1	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.0011	-	0.0054	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		0.465	-	1.09	m.°C/W
Thermal expansion coefficient		3.7	-	6.3	µstrain/°C
Specific heat capacity		810	-	920	J/kg.°C
Melting point		1.23e3	-	1.35e3	°C
Maximum service temperature	*	330	-	380	°C
Minimum service temperature		-273			°C
Flammability		Non-flammable			
Emissivity		0.95			

Hygro-thermal properties

Water absorption		2.5	-	11	%
Frost resistance		Average			

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity	*	1e8	-	1e12	µohm.cm
Dielectric constant (relative permittivity)		10	-	15	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.01	
Dielectric strength (dielectric breakdown)	*	5	-	12	MV/m

Optical properties

Natural color

White when pure, but easily colored by impurities such as iron, to give a red, gray or green hue.

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No			

Acoustic properties

Sound absorption	Average			
Sound isolation	Good			

Processability (scale 1 = impractical to 5 = excellent)

Machinability	2	-	3	

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Unacceptable
Strong acids	Unacceptable
Weak alkalis	Excellent
Strong alkalis	Acceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Limited use
Industrial atmosphere	Unacceptable
Rural atmosphere	Excellent
Marine atmosphere	Acceptable

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.00536	-	0.0059	kg/kg
Embodied energy (virgin grade production)	* 0.161	-	0.178	MJ/kg
Water Usage	* 13	-	14.4	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 18	-	40.9	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO_2-eq)	* 1.18	-	2.68	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Recycle fraction in current supply	1.34	-	1.48	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Limestones vary widely in composition and porosity and their strength and weather resistance reflect this. The softer ones can be sawn and carved by hand, and so are widely used for sculpture and decorative moldings. They weathers in a benign attractive way; the carbonic acid in rain dissolves a very thin surface film over time, making the surface self-cleaning.

Technical notes

Limestone consists mainly of calcite, $CaCO_3$, with sand (SiO_2) and, in some cases magnesium carbonate, $MgCO_3$, (Dolomitic limestone). Limestones vary greatly in hardness and are generally easier to work than sandstones.

Typical uses

Limestone is widely used for building and sculpture. It is calcined in large quantities to make lime (CaO) for cement and as a flux in the smelting of iron.

Tradenames

Portland stone; Bath stone

Related standards

04400 Stone
04410 Rough Stone
04420 Cut Stone
04440 Flagstone
04450 Stone Veneer
04460 Limestone

Links

ProcessUniverse

Reference